Izmir Institute of Technology

Electrical and Electronics Eng. Dep.

EE 471 Modern Software Development Practices and Technologies

In-class exercise Week #7

Monday 09/11/2026

Preliminary Task: Study the shared course materials
[Quickstart, Tensors, Datasets & DataLoaders, Transforms, Build neural network, optimize the model, save&load, Train a small neural network to classify images]

Tasks:

Fashion MNIST

1) Follow the steps in Pytorch quick start tutorial and train a model for garment classification.
2) Modify your cat-kind classifier website you developed last week as a garment classifier website with your brand-new trained model.
3) Containerize your application with cog
4) Apply Hyperparameter optimization to your model
5) Containerize your optimized model with cog
6) Improve your website with a side-by-side comparison of your previous and optimized classifiers.

CIFAR10

1) Apply the same steps of FashionMNIST for CIFAR10 classifier.

Deliverables:

1) Your github repos (with clean source code, Docker File, cog yaml file, and setup instructions).

2) “Task-pre-demo.avi” (duration: max 5 mins). Video of you explaining the basics “Tensors, Datasets & Dataloaders, Transforms, building a neural network to classify images”.
3) “Task-1-demo.avi” Video of you building and running the garment Classifier website. Also in the video, access your running container (e.g., via terminal) and display the live prediction logs directly from inside the container. In addition, demonstrate API Verification (Postman) by sending a POST request with an uploaded image to your Cog container's inference endpoint and receiving the predicted class response.
4) “Task-2-demo.avi” Video of you building and running the CIFAR10 Classifier website. Also in the video, access your running container (e.g., via terminal) and display the live prediction logs directly from inside the container. In addition, demonstrate API Verification (Postman) by sending a POST request with an uploaded image to your Cog container's inference endpoint and receiving the predicted class response.
5) “Tasks-code-explained.avi” (duration: max 3 mins). A screen recording (with audio) of you explaining each code block **relevant to AI-based classification tasks** in your scripts.
6) A max one-page report on the performance of your both classification models (highlight the importance of hyperparameter optimization).